Project: Worked examples
Title: Limits and fits (ISO 286)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the limits of size of a hole and a shaft for a nominal size and ISO 286 tolerance classes, and the resulting fit with its extreme clearances and interferences. It uses the ISO 286-1 formulae, not the tables.

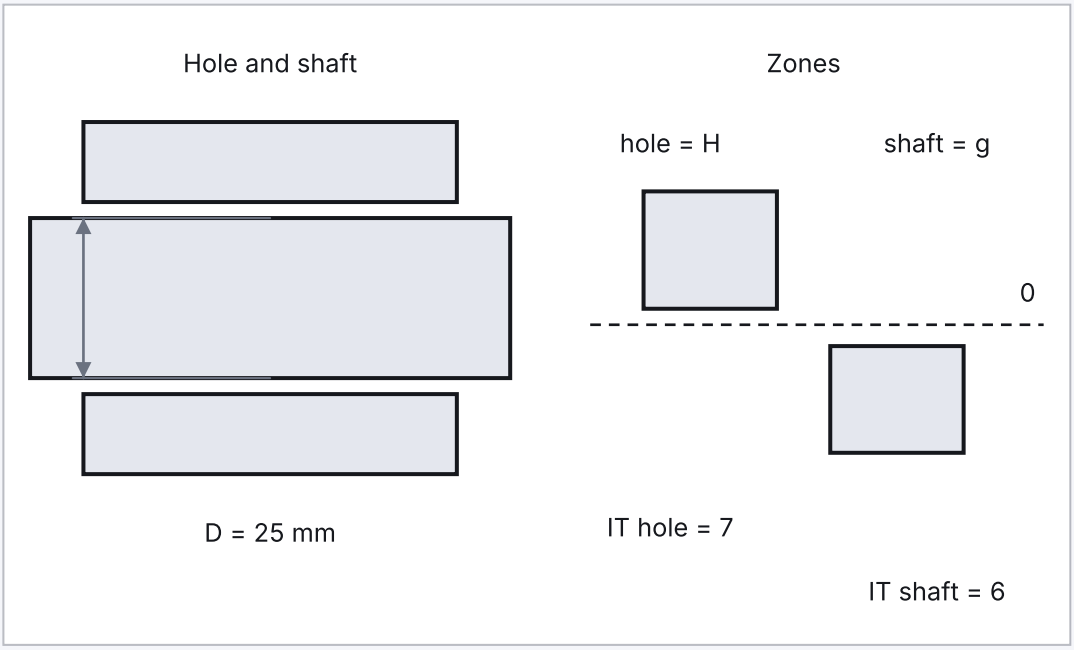


Figure 1 — A hole and its shaft of one nominal size D, and their tolerance zones about the zero line (hole above, shaft below as drawn; the letters place them). Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Standard tolerance factor at the geometric mean D of the size step	i = 0.45·D^(1/3) + 0.001·D (µm)
Width of the tolerance band for the grade	ITn = multiplier × i
Fundamental deviations by letter from the published formulae, and hole deviations by the general rule	$EI_{hole} = -es_{shaft}$ (A–H); $ES_{hole} = -ei_{shaft} + \Delta$ (K–ZC)
Limits of size of the hole and the shaft	limit = nominal + deviation
Fit extremes, the largest and smallest clearance (negative is interference)	$c_{max} = ES_{hole} - ei_{shaft}$; $c_{min} = EI_{hole} - es_{shaft}$
Rolling-bearing bore mode	a bearing bore is not an ISO 286 hole; it has its own tolerance (ISO 492) lying below nominal, so ES_{hole} and EI_{hole} are entered from the bearing record instead of being computed

Basis: Bases of tolerances, deviations and fits — tolerance grades and fundamental deviations[1](§4, §5); Tables of standard tolerance classes and limit deviations[2].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$hole_{kind}$	Hole: an ISO 286 class, or a rolling-bearing bore (ES_{hole} and EI_{hole} from the bearing record, ISO 492)	ISO 286 hole class		default	any
$hole_{letter}$	Hole fundamental deviation letter	H		spec: DRG-SHAFT-12 note 3: H7/g6	any
$hole_{grade}$	Hole IT grade	7		spec: DRG-SHAFT-12 note 3	5 … 11
$shaft_{letter}$	Shaft fundamental deviation letter	g		spec: DRG-SHAFT-12 note 3	any
$shaft_{grade}$	Shaft IT grade	6		spec: DRG-SHAFT-12 note 3	4 … 11
D	Nominal size	25	mm	drawing: DRG-SHAFT-12 rev A	1 … 500 mm

3. Assumptions

A1. IT values and deviations are the unrounded results of the ISO 286-1 formulae at the size step's geometric mean. (The tables apply the standard's own rounding, so tabulated values differ by about 1 µm for the letters a to n.)
A2. For the letters p to zc the formula value can lie several µm below the tabulated deviation. (The tables add their own increments and sub-steps, so p6 at 450 mm is 62.2 µm from the formula against 68 µm tabulated.)
A3. Hole deviations follow the general rule from the shaft deviations, with Δ where the standard applies it. (Δ is ITn − IT(n−1), and the standard's table-specific departures are not reproduced.)

4. Working

[7] D_{step} = size step (lower, upper) [1](§4)
= (18, 30] mm
= 23.238 mm
Geometric mean of the size step

[8] i = 0.45·D^(1/3) + 0.001·D [1](§5)
= D = 23.238 mm
= 1.3074 µm
Standard tolerance unit

[9] IT_{hole} = IT7 [1](§5)
= 16 × i
= 20.918 µm
Hole tolerance

[10] IT_{shaft} = IT6 [1](§5)
= 10 × i
= 13.074 µm
Shaft tolerance

[11] es_{shaft} = fundamental deviation 'g' [1](§5)
= −2.5·D^0.34
= -7.2853 µm
Shaft upper deviation

[12] $ei_{shaft} = es_{shaft} - IT_{shaft}$ [1](§5)
$= -7.2853 - 13.074$
$= -20.359\ \mu m$
Shaft lower deviation
[13] $ES_{hole} = EI_{hole} + IT_{hole}$ [1](§5)
$= 0 + 20.918$
$= 20.918\ \mu m$
Hole upper deviation
[14] EI_{hole} = fundamental deviation 'H' [1](§5)
$= -es_{shaft}$ of the same letter
$= 0\ \mu m$
Hole lower deviation
[15] $D_{hole,max} = D + ES_{hole}$
$= 25\ mm + 20.918\ \mu m$
$= 25.021\ mm$
Hole maximum
[16] $D_{hole,min} = D + EI_{hole}$
$= 25\ mm + 0\ \mu m$
$= 25\ mm$
Hole minimum
[17] $d_{shaft,max} = D + es_{shaft}$
$= 25\ mm + -7.2853\ \mu m$
$= 24.993\ mm$
Shaft maximum
[18] $d_{shaft,min} = D + ei_{shaft}$
$= 25\ mm + -20.359\ \mu m$
$= 24.98\ mm$
Shaft minimum
[19] $c_{max} = ES_{hole} - ei_{shaft}$
$= 20.918\ \mu m - (-20.359\ \mu m)$
$= 41.277\ \mu m$
Maximum clearance (negative: minimum interference)
[20] $c_{min} = EI_{hole} - es_{shaft}$
$= 0\ \mu m - (-7.2853\ \mu m)$
$= 7.2853\ \mu m$
Minimum clearance (negative: maximum interference)
[21] $i_{max} = es_{shaft} - EI_{hole}$
$= (-7.2853\ \mu m) - 0\ \mu m$
$= -7.2853\ \mu m$
Maximum interference
[22] $i_{min} = ei_{shaft} - ES_{hole}$
$= (-20.359\ \mu m) - 20.918\ \mu m$
$= -41.277\ \mu m$
Minimum interference (negative: maximum clearance)
[23] fit = fit class
$=$ H7/g6
$=$ "clearance"
Fit kind: clearance, transition or interference
[24] $c_{mean} = (c_{max} + c_{min}) / 2$
$= (41.277\ \mu m + 7.2853\ \mu m) / 2$
$= 24.281\ \mu m$
Mean clearance

5. Results

SYMBOL	VALUE	DESCRIPTION
c_{max}	41.28 µm	Maximum clearance
c_{min}	7.285 µm	Minimum clearance
i_{max}	-7.285 µm	Maximum interference
i_{min}	-41.28 µm	Minimum interference
ES_{hole}	20.92 µm	Hole upper deviation
EI_{hole}	0 µm	Hole lower deviation
es_{shaft}	-7.285 µm	Shaft upper deviation
ei_{shaft}	-20.36 µm	Shaft lower deviation
IT_{hole}	20.92 µm	Hole tolerance
IT_{shaft}	13.07 µm	Shaft tolerance
$D_{hole,max}$	25.02 mm	Hole maximum
$D_{hole,min}$	25 mm	Hole minimum
$d_{shaft,max}$	24.99 mm	Shaft maximum
$d_{shaft,min}$	24.98 mm	Shaft minimum
fit	"clearance"	Fit kind
c_{mean}	24.28 µm	Mean clearance

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Fit classes against the fit kind	H7/g6	clearance	–	

7. Applicability

None.

Defaults used (inputs left empty):
$hole_{kind}$: not entered, so the default of 'ISO 286 hole class' is used.

8. Verification

Template mech.iso286-fit@1.6.1 Content hash 499f772f7d32918a Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 1: Basis of Tolerances, Deviations and Fits. ISO 286-1:2010. ISO; 2010.
2. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 2: Tables of Standard Tolerance Classes and Limit Deviations for Holes and Shafts. ISO 286-2:2010. ISO; 2010.